

Low-Light Image Enhancement Using Pixel-Adaptive Gamma Correction with Practical Denoising, Auto-Tuning, and Acceleration

Project Report
Low-Light Image Enhancement Study

Abstract—Low-light image enhancement is important for surveillance, mobile imaging, and downstream computer vision because underexposed scenes usually contain poor contrast, limited detail visibility, and amplified sensor noise. This project implements a low-light enhancement pipeline derived from pixel-adaptive gamma correction and a dehazing-inspired recovery model. The implementation was further improved with three practical enhancements: stronger denoising, automatic per-image parameter tuning, and a faster processing path for larger images. Evaluation on the 49 low-light images bundled with the repository shows consistent visual improvement and strong gains in no-reference image quality indicators. Mean grayscale brightness increased from 15.468 to 109.333, mean contrast increased from 10.791 to 53.721, and mean entropy increased from 4.904 to 7.405. The average runtime of the upgraded implementation on the local dataset was 0.625 s per image. Qualitative examples and dataset-level plots indicate that the modified pipeline improves visibility while preserving more detail by recovering the final image from the original input rather than from the denoised guide image.

Index Terms—low-light enhancement, gamma correction, transmission map, denoising, adaptive parameter tuning, image restoration, dehazing prior

I. INTRODUCTION

Low-light images often suffer from low brightness, compressed dynamic range, color distortion, and visible noise. A practical enhancement method must brighten dark regions without washing out highlights or destroying useful texture. The base project in this repository approaches the problem by transforming low-light enhancement into a dehazing-style recovery task. It estimates atmospheric light from the inverted image, applies pixel-adaptive gamma correction, computes a transmission map, and reconstructs the enhanced result.

The original method is effective, but three implementation-level improvements were especially useful for practical deployment. First, noise removal needed strengthening because dark scenes frequently contain strong luminance and chroma noise. Second, fixed parameters such as `gamma_max` are not ideal for images with very different exposure and contrast characteristics. Third, the runtime path could be simplified and partially downscaled to improve throughput on larger datasets.

This paper documents the upgraded implementation and summarizes qualitative and quantitative results on the repository dataset.

II. RELATED BASIS

The project is based on the low-light enhancement formulation introduced by Jeon *et al.* [1], where gamma correction

priors are integrated with mixed color-space reasoning. The implementation also uses a dehazing-style atmospheric light and transmission estimation perspective related to the dark channel prior of He *et al.* [2]. In this work, those ideas are retained while the engineering pipeline is improved for stronger robustness and easier batch use.

III. METHODOLOGY

A. Baseline Pipeline

Let I denote the low-light input image, A the estimated atmospheric light, and t the transmission map. The baseline enhancement path follows a dehazing-inspired recovery model after inverting and normalizing the image. Pixel-adaptive gamma correction is applied to the normalized guide image, and the corrected representation is used to estimate the transmission map for final recovery.

B. Stronger Noise Removal

The denoising stage was upgraded to improve stability in atmospheric light and transmission estimation.

- 1) Noise level is estimated from the median absolute Laplacian response on the grayscale input.
- 2) `fastNlMeansDenoisingColored` is auto-tuned according to scene darkness and estimated noise.
- 3) A bilateral filtering stage is added for especially noisy or dark scenes.
- 4) For large images, the denoising prior can be estimated at reduced resolution and blended with an edge-preserving result.

After visual review, an additional correction was applied: the final reconstruction uses the original input rather than the denoised image. This keeps denoising in the estimation path only and avoids an over-smoothed appearance in the final output.

C. Automatic Parameter Tuning

The implementation no longer treats `gamma_max` as a fixed global constant. Instead, the user-specified value is used as an upper bound, and an image-specific effective gamma limit is derived from three cues: mean brightness, contrast, and estimated noise level. Darker and flatter images receive stronger enhancement, while relatively brighter inputs receive less aggressive correction. Across the repository dataset, the

tuned gamma ranged from 5.279 to 6.000, with a mean of 5.749.

D. Faster Processing Path

Three runtime improvements were introduced.

- 1) Python channel loops were replaced with vectorized NumPy expressions.
- 2) `skimage` morphology in the dark-channel step was replaced with `cv2.erode`.
- 3) Only the expensive estimation branch is downsampled for large images, after which the transmission and gamma maps are resized back to full resolution.

These changes preserve the original method structure while improving throughput and reducing implementation overhead.

IV. EXPERIMENTAL SETUP

A. Dataset

The repository folder `input/` contains 49 low-light images used for this study.

B. Environment

The implementation uses Python with OpenCV, NumPy, and Matplotlib. All analysis figures and tables were generated from the project outputs using the script `generate_ieee_report_assets.py`.

C. Evaluation Protocol

Because no paired ground-truth bright images are included in the repository, evaluation was performed with no-reference indicators and visual inspection.

- 1) Brightness: mean grayscale intensity.
- 2) Contrast: grayscale standard deviation.
- 3) Entropy: information content from the grayscale histogram.
- 4) Sharpness: variance of the Laplacian.
- 5) Runtime: wall-clock runtime reported by the enhancement function.

Per-image measurements are stored in `report_assets/tables/dataset_metrics.csv`.

V. QUALITATIVE RESULTS

Fig. 1 shows the demo visualization generated by the project demo script. It includes the original image, enhanced result, adaptive gamma map, histograms, and the transmission map.

Fig. 2 shows representative original and enhanced comparisons. The selected examples span darker and relatively brighter low-light cases, demonstrating that the auto-tuned pipeline adapts across the dataset.

Qualitatively, the upgraded pipeline consistently increases scene visibility, expands local and global contrast, and assigns stronger gamma correction to darker regions. The decision to reconstruct from the original input rather than the denoised guide image also preserves a more natural level of detail.

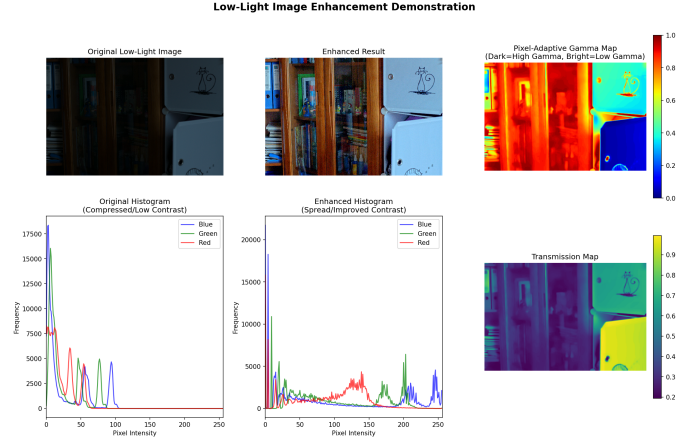


Fig. 1. Demo visualization generated by the project demo script.

TABLE I
AVERAGE NO-REFERENCE METRICS OVER THE DATASET

Metric	Original Mean	Enhanced Mean	Delta
Brightness	15.468	109.333	93.865
Contrast	10.791	53.721	42.930
Entropy	4.904	7.405	2.501
Sharpness	58.974	5706.727	5647.752

VI. QUANTITATIVE RESULTS

Fig. 3 summarizes average dataset-level quality metrics before and after enhancement. Fig. 4 shows how the auto-tuned gamma changes with input brightness and how runtime scales with image size.

Table I reports the average no-reference metrics over all 49 images.

Table II reports representative per-image results for four selected samples.

The average runtime was 0.625 s per image, with a minimum of 0.529 s and a maximum of 0.682 s on the evaluation set.

VII. DISCUSSION

The results indicate that the project enhancements improve both practicality and output quality.

- 1) The stronger denoising stage stabilizes light estimation in dark scenes.
- 2) Automatic gamma tuning removes the need to manually choose a single `gamma_max` for every image.
- 3) Vectorization and reduced-resolution estimation improve efficiency without changing the overall algorithm design.
- 4) Reconstructing the final result from the original input avoids unnecessary smoothing and yields a more natural appearance.

There are also several limitations. First, the metrics are no-reference and therefore do not directly measure fidelity against ground truth. Second, the strong increase in Laplacian variance partly reflects stronger edges and contrast, so it should be interpreted together with visual inspection. Third,

TABLE II
REPRESENTATIVE SAMPLE RESULTS

Image	Bright. Gain	Contr. Gain	Entropy Gain	Gamma	Runtime
2.png	71.42	38.04	1.97	5.28	0.604
28.png	75.42	44.69	3.65	6.00	0.529
63.png	93.63	40.31	2.67	5.89	0.679
13.png	133.27	40.83	2.28	5.77	0.638

the repository dataset is relatively small and contains similarly sized images, which limits runtime variability.

VIII. CONCLUSION

This project demonstrates a practical low-light enhancement system built on pixel-adaptive gamma correction and dehazing-inspired recovery. Three targeted implementation upgrades were added: stronger denoising, automatic parameter tuning, and faster processing. Evaluation on the repository dataset shows large gains in brightness, contrast, and entropy while maintaining visually stronger detail by recovering the final image from the original low-light input. The resulting codebase is more consistent, easier to analyze, and better suited for both demonstration and batch-processing workflows.

ACKNOWLEDGMENT

The report is based on the repository implementation and references the original low-light enhancement paper of Jeon *et al.* [1].

REFERENCES

- [1] J. J. Jeon, J. Y. Park, and I. K. Eom, "Low-light image enhancement using gamma correction prior in mixed color spaces," *Pattern Recognition*, 2023. [Online]. Available: <https://doi.org/10.1016/j.patcog.2023.110001>
- [2] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 33, no. 12, pp. 2341–2353, 2011.

Representative Low-Light Enhancement Results

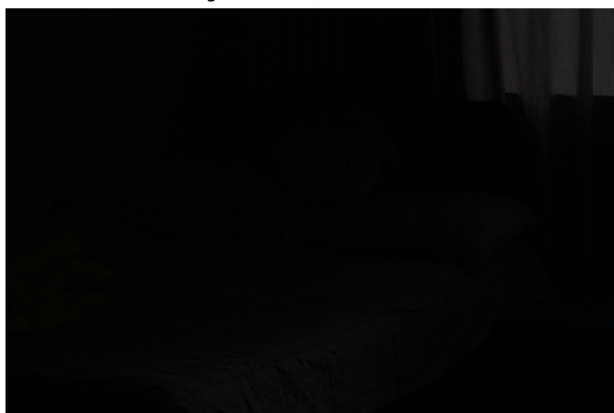
Original 2.png
Brightness 24.6, Contrast 23.2



Enhanced 2.png
Brightness 96.0, Contrast 61.3



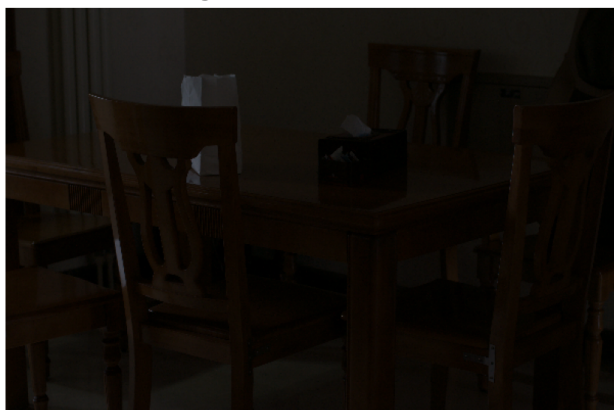
Original 28.png
Brightness 4.3, Contrast 5.9



Enhanced 28.png
Brightness 79.7, Contrast 50.6



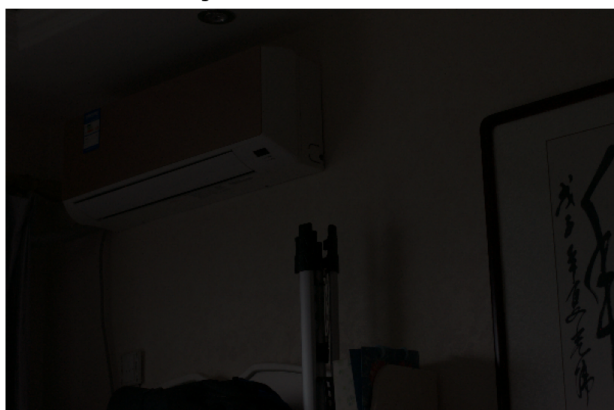
Original 63.png
Brightness 12.8, Contrast 8.8



Enhanced 63.png
Brightness 106.4, Contrast 49.1



Original 13.png
Brightness 18.4, Contrast 8.3



Enhanced 13.png
Brightness 151.6, Contrast 49.1



Fig. 2. Representative original and enhanced image comparisons from the dataset.



Fig. 3. Dataset-level metric comparison for original and enhanced images.

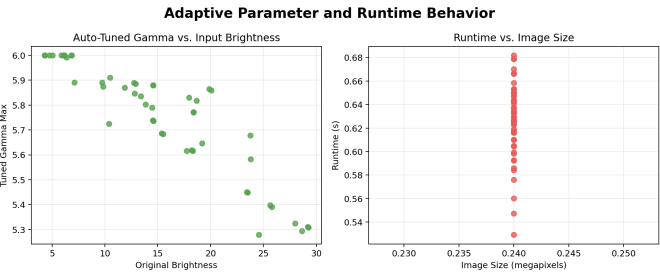


Fig. 4. Adaptive gamma and runtime behavior across the dataset.